

MH371 truth-blind cruise control

wide_mach_altitude_loose_bfo | 2 deterministic seeds | generated from run artifacts

Control result

MODEL FAMILY	PARTICLES
wide_mach_altitude_loose_bfo	12000 x 2 seeds
TERMINAL MEAN ERROR	TERMINAL MASS WITHIN 100 NM
86.7 to 89.1 NM	35.5 to 48.3%
TERMINAL ROOT ESS	BFO OBSERVATION SD
28.1 to 29.9	7.0 Hz
TRUTH-IMPLIED BFO BIAS	BTO EQUATION CLOSURE
167.2 to 171.6 Hz	3.076e-1 us max

Evidence and inference conditions

- Cruise only: 01:48 UTC initial state through the selected 06:48:33 UTC SATCOM observation.
- Later ACARS positions, trajectory, Mach, altitude, and headings were excluded from inference and loaded only here.
- ERA5 temperature and wind plus IGRF-14 east-positive magnetic declination drive every propagation step.
- Satellite oscillator and Perth GES AFC terms remain separate inputs; aircraft BFO bias is analytically estimated.
- Each map combines deterministic seed replicates with equal replicate weight.

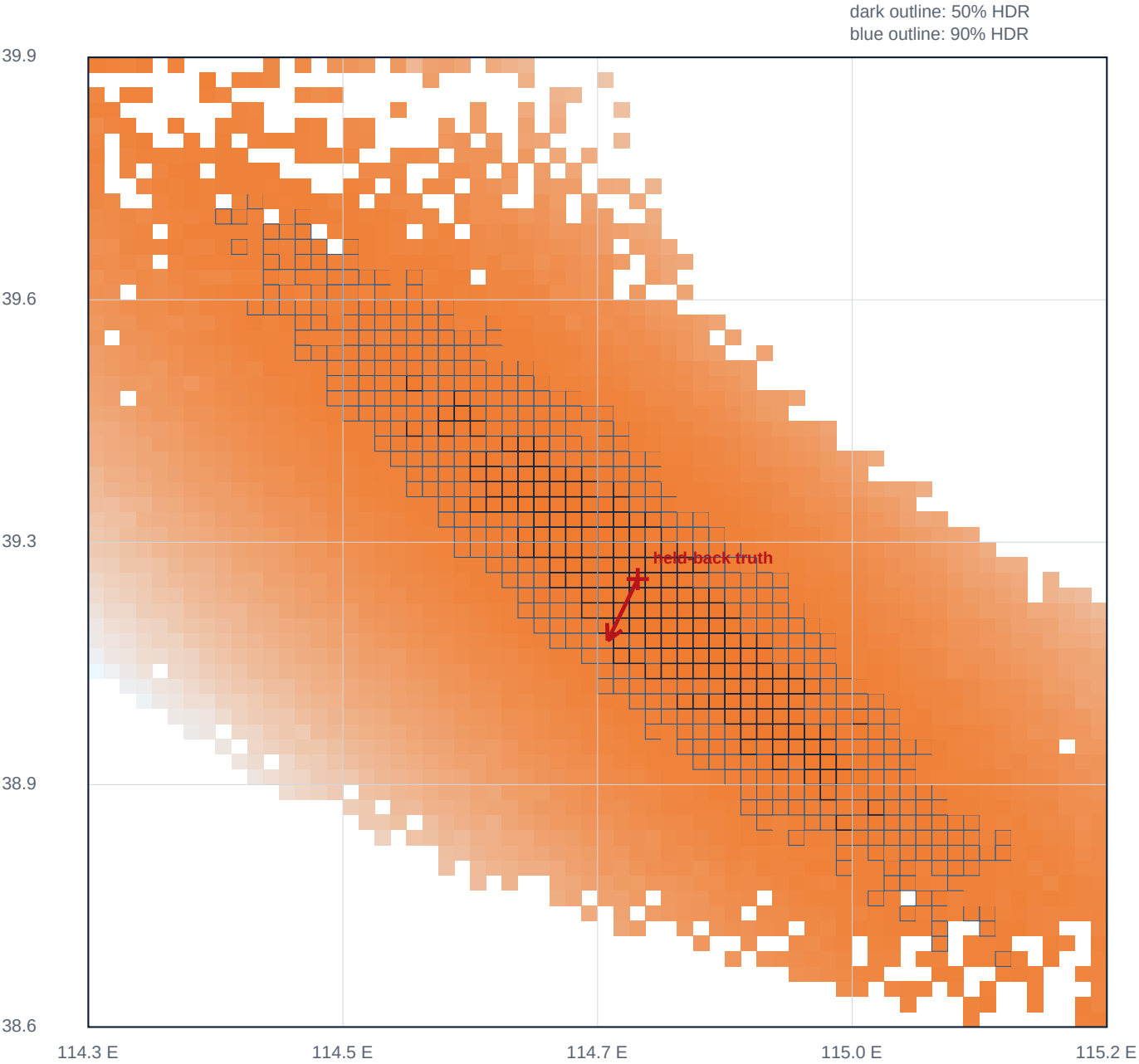
Material limitations

- held-back ACARS positions and trajectories were loaded only by this scorer
- posterior localization is conditional on the declared motion and BFO-noise family
- this known-flight control does not establish adequacy for the MH370 accident flight

The following 6 pages show posterior density, held-back position, and true-heading arrow at each selected SATCOM arc.

mh371_0200 SATCOM arc

2014-03-07T01:56:47.921Z | truth 39.2204 deg, 114.7725 deg | true heading 206.0 deg



Posterior and control metrics

MEAN POSITION ERROR
0.2 NM

MACH: POSTERIOR / TRUTH
0.756 / 0.759

GROUND SPEED: POSTERIOR / TRUTH
399 / 383 kt

TRUTH-IMPLIED BFO BIAS
167.2 Hz

POSTERIOR WITHIN 100 NM
100.0%

ALTITUDE: POSTERIOR / TRUTH
27982 / 27599 ft

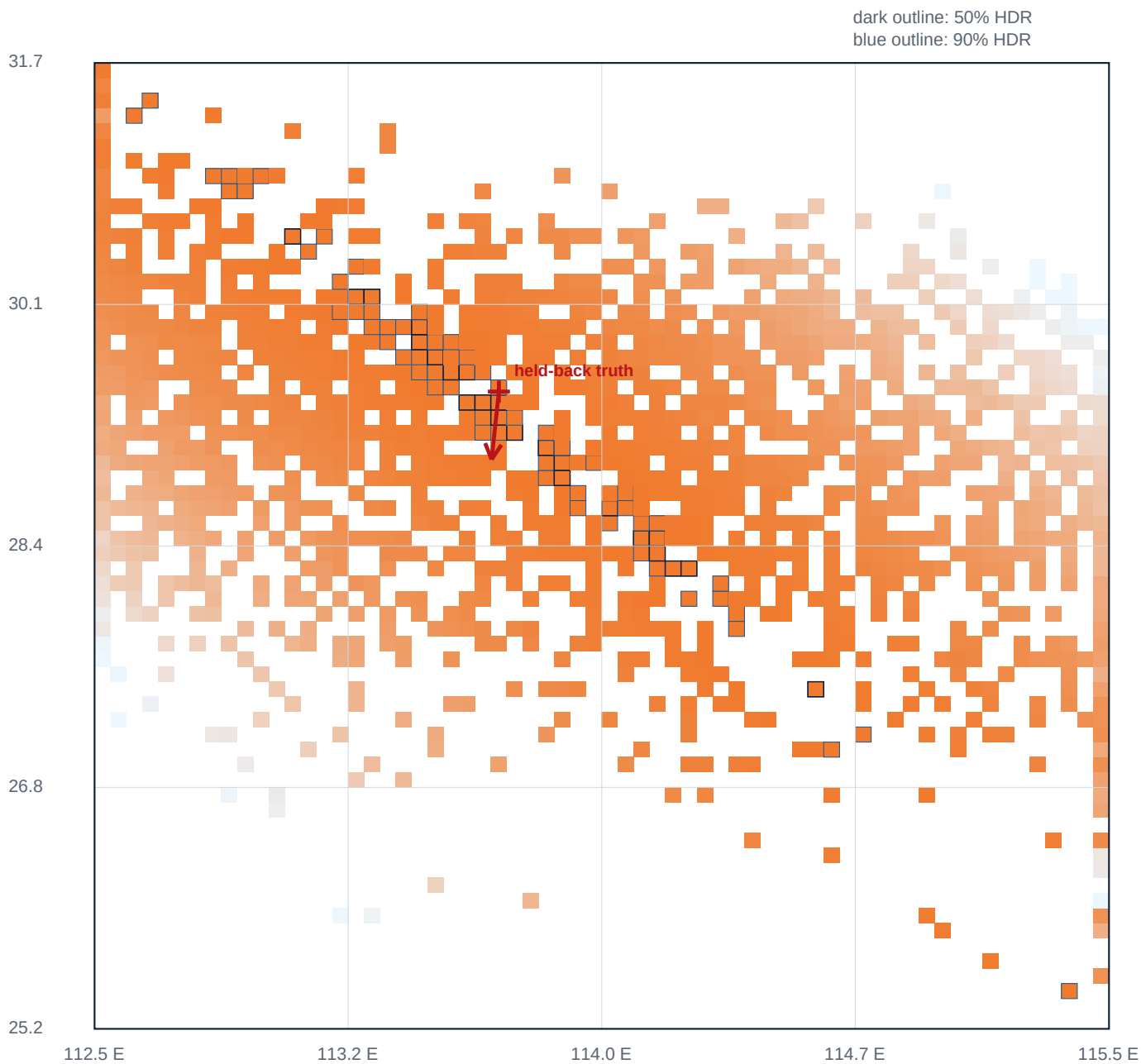
LATENT BFO BIAS
166.0 Hz

PARTICLE / ROOT ESS
4466 / 4465.9

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0300 SATCOM arc

2014-03-07T03:20:22.428Z | truth 29.4730 deg, 113.6856 deg | true heading 186.1 deg



Posterior and control metrics

MEAN POSITION ERROR

16.2 NM

MACH: POSTERIOR / TRUTH

0.757 / 0.753

GROUND SPEED: POSTERIOR / TRUTH

431 / 452 kt

TRUTH-IMPLIED BFO BIAS

171.6 Hz

POSTERIOR WITHIN 100 NM

90.9%

ALTITUDE: POSTERIOR / TRUTH

29203 / 28368 ft

LATENT BFO BIAS

168.3 Hz

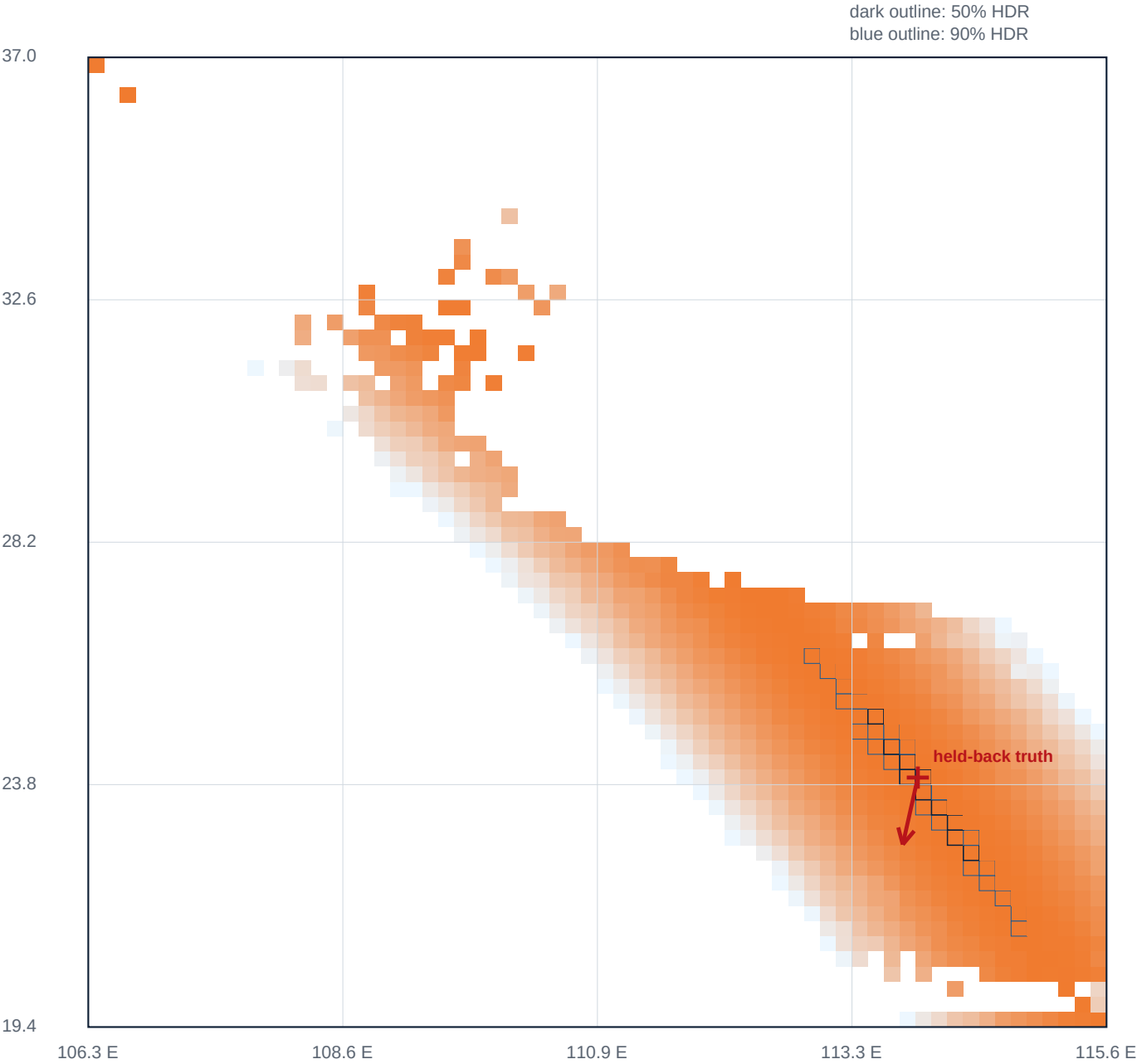
PARTICLE / ROOT ESS

67 / 65.8

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0400 SATCOM arc

2014-03-07T04:04:57.916Z | truth 23.9533 deg, 113.8817 deg | true heading 192.7 deg



Posterior and control metrics

MEAN POSITION ERROR
12.8 NM

MACH: POSTERIOR / TRUTH
0.756 / 0.832

GROUND SPEED: POSTERIOR / TRUTH
438 / 455 kt

TRUTH-IMPLIED BFO BIAS
169.6 Hz

POSTERIOR WITHIN 100 NM
78.6%

ALTITUDE: POSTERIOR / TRUTH
30311 / 38100 ft

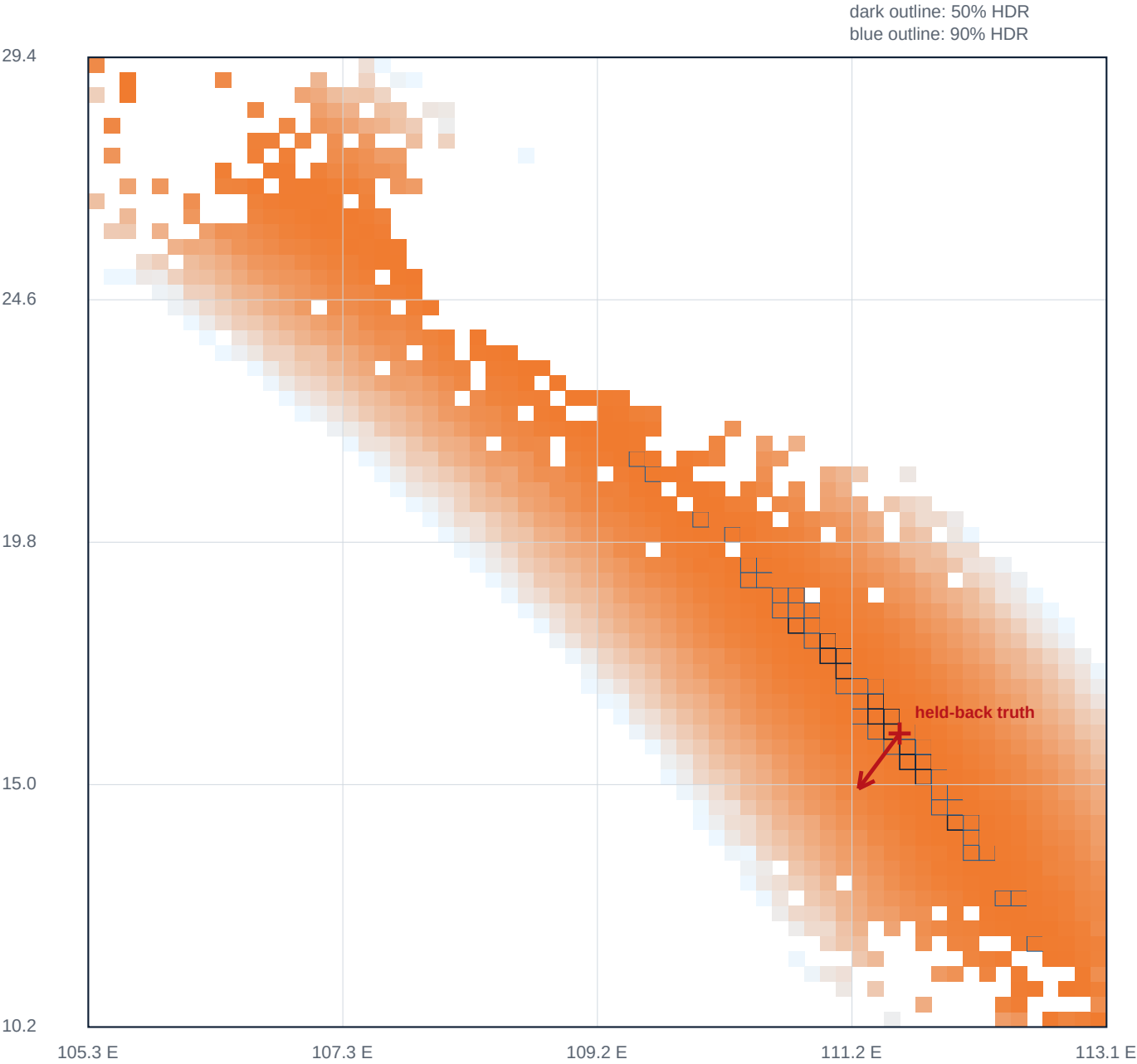
LATENT BFO BIAS
169.0 Hz

PARTICLE / ROOT ESS
294 / 53.6

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0500 SATCOM arc

2014-03-07T05:11:37.411Z | truth 15.9956 deg, 111.5330 deg | true heading 216.7 deg



Posterior and control metrics

MEAN POSITION ERROR
48.0 NM

MACH: POSTERIOR / TRUTH
0.746 / 0.830

GROUND SPEED: POSTERIOR / TRUTH
435 / 462 kt

TRUTH-IMPLIED BFO BIAS
169.7 Hz

POSTERIOR WITHIN 100 NM
59.6%

ALTITUDE: POSTERIOR / TRUTH
30906 / 39998 ft

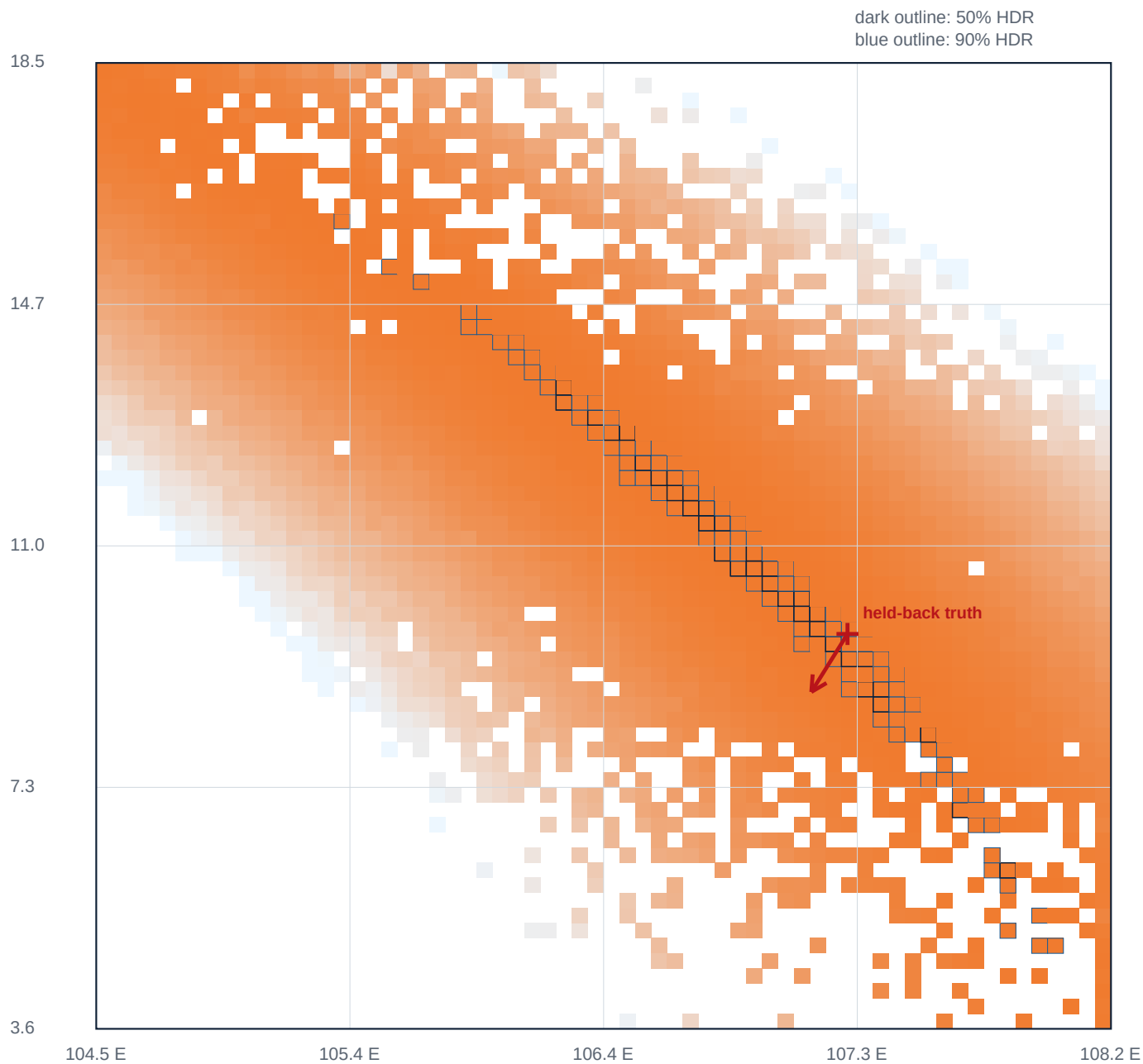
LATENT BFO BIAS
169.3 Hz

PARTICLE / ROOT ESS
179 / 43.5

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0600 SATCOM arc

2014-03-07T06:09:44.409Z | truth 9.6835 deg, 107.2620 deg | true heading 212.0 deg



Posterior and control metrics

MEAN POSITION ERROR

65.9 NM

MACH: POSTERIOR / TRUTH

0.754 / 0.813

GROUND SPEED: POSTERIOR / TRUTH

455 / 474 kt

TRUTH-IMPLIED BFO BIAS

170.6 Hz

POSTERIOR WITHIN 100 NM

46.7%

ALTITUDE: POSTERIOR / TRUTH

30043 / 39997 ft

LATENT BFO BIAS

170.2 Hz

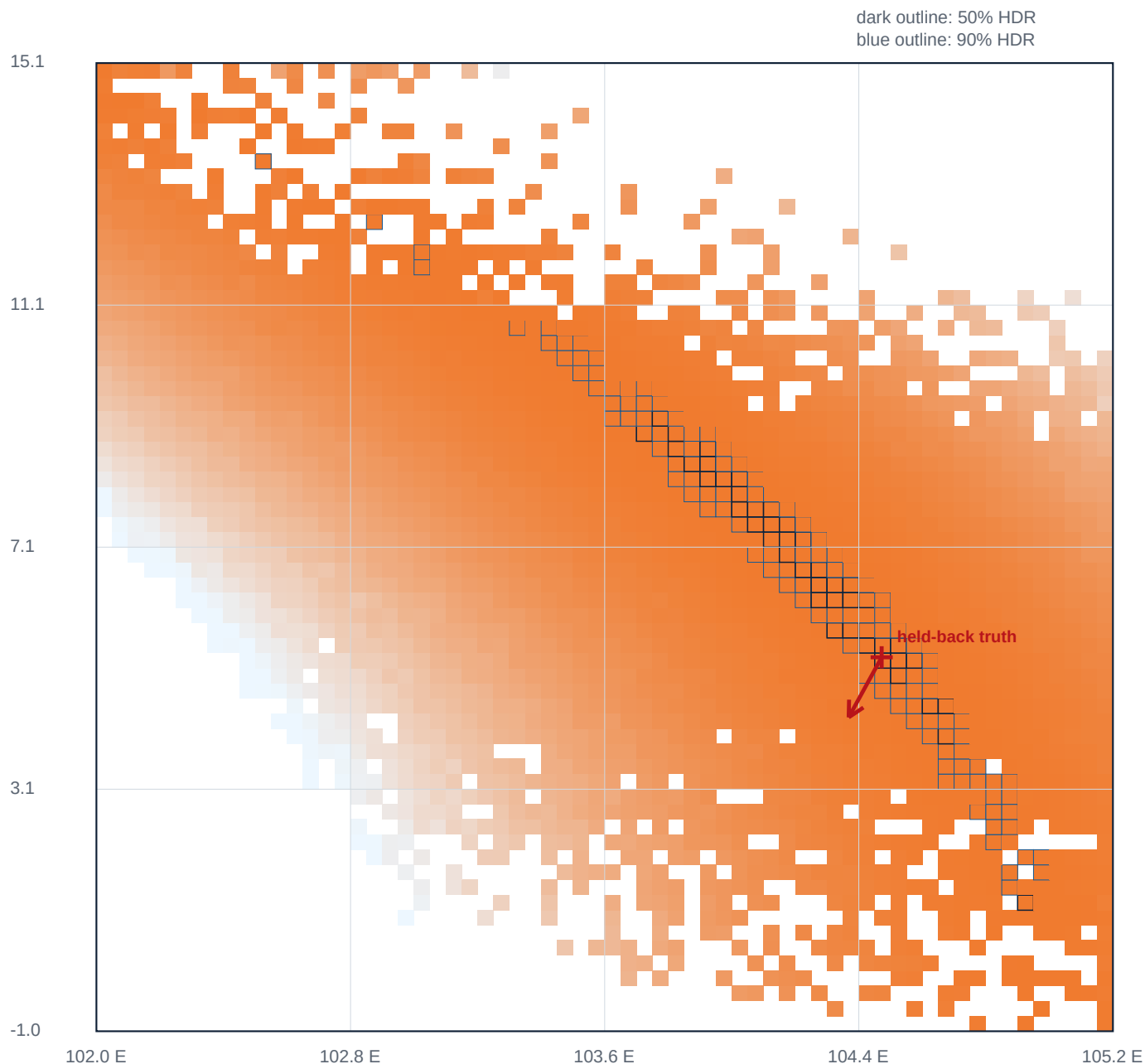
PARTICLE / ROOT ESS

272 / 33.1

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0648 SATCOM arc

2014-03-07T06:48:33.907Z | truth 5.2504 deg, 104.4874 deg | true heading 208.8 deg



Posterior and control metrics

MEAN POSITION ERROR

87.9 NM

MACH: POSTERIOR / TRUTH

0.749 / 0.821

GROUND SPEED: POSTERIOR / TRUTH

460 / 483 kt

TRUTH-IMPLIED BFO BIAS

168.6 Hz

POSTERIOR WITHIN 100 NM

41.9%

ALTITUDE: POSTERIOR / TRUTH

29736 / 40000 ft

LATENT BFO BIAS

170.6 Hz

PARTICLE / ROOT ESS

489 / 29.0

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.